# Kolmogorov Smirnov - Examples in R

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Here, we explore one and two-sample Kolmogorov Smirnov (KS) tests.

### Simple Example

X and Y were both generated from a (continuous) Uni(0,1) distribution - uniform on 0 to 1.

```
x <- c(0.28, 0.39, 0.69, 0.86)
y <- c(0.32, 0.57, 0.65, 0.81, 0.92)
ks.test(x, y) # function is in base stats package
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  x and y
## D = 0.3, p-value = 0.9683
## alternative hypothesis: two-sided
```

What is your conclusion from this test procedure?

### An example for simulated data from different uniform distributions.

Now, U is from a Uniform(0,1) distribution, but V is from a Uniform(0, 10) distribution.

```
u <- c(0.73, .34, .35, .02, .56) # generated from a Uniform(0, 1) distribution
v <- c(7.61, 2.49, 2.02, 4.79, 2.26) # generated from a Uniform(0, 10) distribution
ks.test(u, v)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  u and v
## D = 1, p-value = 0.007937
## alternative hypothesis: two-sided
```

Did the test procedure detect the difference in the two distributions?

### Two Sample Examples

The machine repair example we saw previously in the two-sample setting concluded that there was not a difference in center. For this data, the center is known to NOT change, but the repair was supposed to fix an issue with the variance. (That would also mean the shift model wouldn't apply - it's hard to tell with this small a sample size, but if you KNEW the issue was with the variance, you would have ruled the shift model out.) Thus, we'd like for the two samples (before and after the repair) to be from different distributions (same center, different variance/spreads).

An automatic refill machine (16 oz expected) just underwent repairs. Was there any difference in the before/after oz delivered?

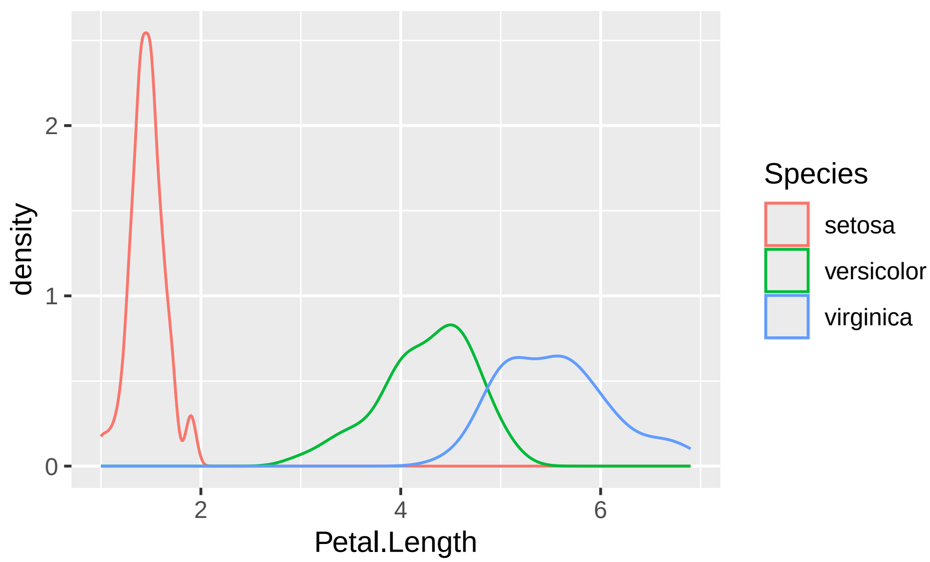
```r
before <- c(16.61, 16.22, 15.34, 15.88, 16.05)
after <- c(15.81, 15.89, 16.02, 16.19, 16.06)
ks.test(before, after)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  before and after
## D = 0.4, p-value = 0.873
## alternative hypothesis: two-sided
```

Unfortunately, the test doesn't provide the result we want. Here, since there is knowledge that the change should be in the variance, you should actually use a test for that - which is the method discussed in the early sections of chapter 5 in Hollander, Wolfe, and Chicken. Feel free to explore those sections.

So, what kind of distributional changes CAN we detect? (Larger sample sizes help too.)

```r
data(iris)
setosa <- filter(iris, Species == "setosa")
virginica <- filter(iris, Species == "virginica")
versicolor <- filter(iris, Species == "versicolor")
ggplot(data = iris, aes(x = Petal.Length, color = Species)) + geom_density()
```




It should be pretty clear that ALL three of these distributions are different. So, the KS tests should confirm that. For example, the distribution of Petal.Length for setosa is entirely below that of the other two, so that should be a large difference in empirical CDFs and then an extremely tiny p-value:

```r
ks.test(setosa$Petal.Length, virginica$Petal.Length)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  setosa$Petal.Length and virginica$Petal.Length
## D = 1, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

Note that if you see a p-value of < 2.2e-16, that means it's smaller than the smallest number R can report.

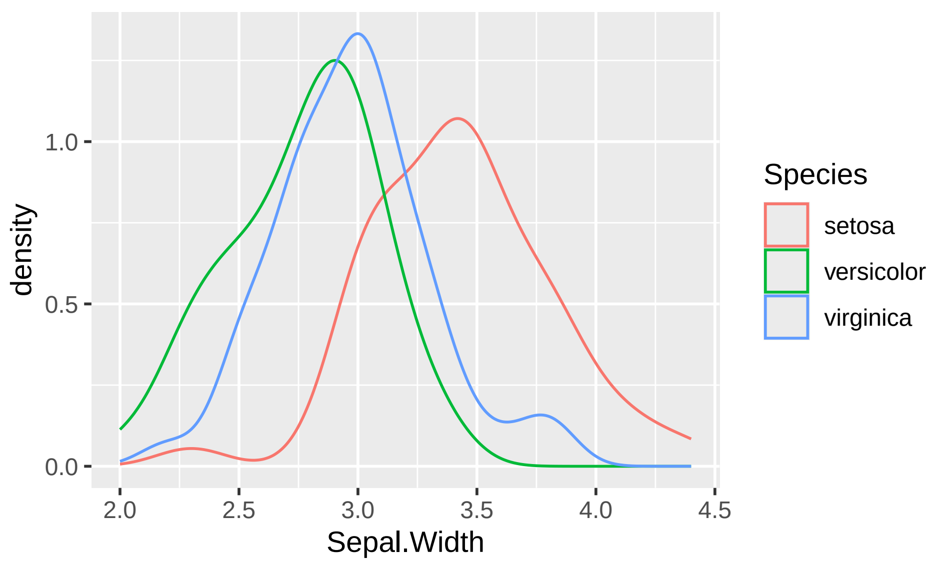
There are alternative ways to run the test - if you didn't split the data set up as we did, and wanted to run it directly from the iris data set. The same result can be obtained with:

```
with(iris, ks.test(Petal.Length[Species == "setosa"],
                   Petal.Length[Species == "virginica"]))
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  Petal.Length[Species == "setosa"] and Petal.Length[Species == "virginica"]
## D = 1, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

If we swap to Sepal.Width, the results are more interesting.

```
ggplot(data = iris, aes(x = Sepal.Width, color = Species)) + geom_density()
```




The setosa distribution for Sepal.Width looks different from the other two, shifted up a bit with some additional spread, but the other two distributions are very similar.

```
# looks similar but virginica shifted up a bit
ks.test(versicolor$Sepal.Width, virginica$Sepal.Width)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  versicolor$Sepal.Width and virginica$Sepal.Width
## D = 0.26, p-value = 0.02826
## alternative hypothesis: two-sided
```

```
ks.test(versicolor$Sepal.Width, setosa$Sepal.Width) # setosa is above
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  versicolor$Sepal.Width and setosa$Sepal.Width
## D = 0.68, p-value = 6.437e-12
## alternative hypothesis: two-sided
```

```r
ks.test(setosa$Sepal.Width, virginica$Sepal.Width) # setosa is above
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  setosa$Sepal.Width and virginica$Sepal.Width
## D = 0.52, p-value = 2.495e-07
## alternative hypothesis: two-sided
```

What do the results suggest at a significance level of 0.01 for each test? (This is to control the overall alpha - we don't want to do each test at a 0.05 level). Are you surprised by the results?

## One Sample Examples

You can use KS tests to check to see if a particular sample has a particular distribution (i.e. normal, t, F, gamma, beta, etc. distribution… whatever distribution you want). We are often particularly interested in checking for normal distributions, because that is usually related to the condition that differs between parametric and nonparametric methods. If the condition does not hold for the appropriate values (which could be variables or a difference in variables), we swap to our nonparametric methods.

Here, we show you code that lets you check for normal distributions with some specified mean and standard deviation.

```r
X <- rnorm(100, 10, 1) # generates 100 observations from a N(10,1)
ks.test(X, "pnorm", 10, 1) # provide X, dist, mean, SD
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  X
## D = 0.10636, p-value = 0.208
## alternative hypothesis: two-sided
```

```r
ks.test(X, "pnorm") # if mean, SD not provided, the defaults are used - 0 and 1 resp.
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  X
## D = 1, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

```r
ks.test(X, "pnorm", 9.5, 2) # values slightly off from simulation
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  X
## D = 0.28321, p-value = 2.158e-07
## alternative hypothesis: two-sided
```

When applying this to actual data for a particular variable, you will need to estimate the mean and SD using sample statistics - these are the best estimates of those values, and so you'd use them as your guesses here for the model parameters. For example, if we want to see if the virginica Sepal.Width distribution is normal, we would use:

```r
favstats(~ Sepal.Width, data = virginica) # pull out mean and SD
```

```
##   min  Q1 median    Q3 max  mean        sd  n missing
##   2.2 2.8      3 3.175 3.8 2.974 0.3224966 50       0
```

```r
mean(~ Sepal.Width, data = virginica) # could also pull out this way and save
```

```
## [1] 2.974
```

```r
sd(~ Sepal.Width, data = virginica)
```

```
## [1] 0.3224966
```

```r
ks.test(virginica$Sepal.Width, "pnorm", 2.974, 0.3225)
```

```
## Warning in ks.test.default(virginica$Sepal.Width, "pnorm", 2.974, 0.3225): ties
## should not be present for the one-sample Kolmogorov-Smirnov test
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  virginica$Sepal.Width
## D = 0.12787, p-value = 0.387
## alternative hypothesis: two-sided
```

If you want to fit other distributions, let your instructor know so they can help you with the commands and with providing estimates of the model parameters. A uniform distribution is fairly easy to try - you'd use punif and need to provide the start and end points (so use min and max roughly (there are slightly better estimates to use, but that's a reasonable starting point)).

Note that the procedure will give you a message about ties.

## Data Sources

Machine repair data was simulated.

Iris data. Available in R with data(iris). Original sources referenced in R help:

Fisher, R. A. (1936) The use of multiple measurements in taxonomic problems. Annals of Eugenics, 7, Part II, 179–188. doi:10.1111/j.1469-1809.1936.tb02137.x.

Anderson, Edgar (1935). The irises of the Gaspe Peninsula, Bulletin of the American Iris Society, 59, 2–5.

## Standard Reference

These materials were created for use in an undergraduate nonparametrics course. Where provided, textbook references refer to Nonparametric Statistical Methods (3rd Edition) by Hollander, Wolfe, and Chicken. Notation is chosen to be consistent with that text. However, materials may be used independently of the textbook.

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